

ITALIAN SCHOOL MEALS OBSERVATORY

Pilot 2026

Cross-municipality assessment of menu composition,
nutritional transparency and school food-system governance

Seven municipalities • 170 coded lunches • Codebook v1.0 • Equal-depth governance review

Technical report — Public-document phase v1.0 — September 2026

Prepared by Vito Leone. Independent descriptive desk study based on publicly available municipal, health-authority and procurement documentation. The pilot is not nationally representative and does not rank municipalities.

Executive summary

This pilot developed and tested a reproducible framework for comparing Italian school-meal systems using public documents. The study combined day-level menu coding with a standardized review of governance, monitoring, procurement, equity and outcome transparency across Torino, Firenze, Napoli, Roma, Guspini, Cittadella and Argenta.

A total of 170 scheduled school lunches were coded using a frozen operational codebook. Indicators covered vegetables, fresh fruit, substantial legume dishes, fish, fresh and processed meat, eggs, cheese, wholegrain items, fried/breaded items and plant-protein main dishes. Missing public information was retained as missing rather than converted to zero. A stratified random 10% quality audit (17/170 rows) was then performed. The audit detected a Roma source-version issue and triggered a complete 45-row re-code against the updated official winter menu; post-correction classification agreement was 100%.

Across the seven cases, basic menu rotation was more convergent than governance and transparency. Fish was close to one serving per five school lunches in every municipality; legumes generally clustered around one to 1.5 weekly exposures. Vegetables were also consistently frequent. Processed-meat exposure varied more, ranging from approximately 0.44 to 3.0 exposures per 20 lunches in the coded seasonal cycles. These values describe published menu provision, not consumption or nutritional adequacy.

The strongest cross-case finding is an outcome-transparency gap. Menus, procurement specifications, special-diet rules and monitoring procedures are often public, whereas quantitative results on plate waste, dish acceptance, non-conformities, penalties and achieved sustainable-procurement shares are much less consistently available. Torino, Firenze and Argenta provide useful quantitative user or administrative outcomes; Napoli mainly publishes qualitative outcomes; Roma, Guspini and Cittadella document monitoring architectures but leave important current outcome datasets unpublished or difficult to locate.

The pilot therefore supports two conclusions. First, a public-document school-food observatory is technically feasible and can produce a traceable cross-municipality dataset. Second, the next research value lies less in adding more menus and more in obtaining existing implementation datasets from municipalities and contractors. The pilot is intended as a portfolio and methods demonstration, not as a national league table or legal-compliance assessment.

Key pilot numbers

Indicator	Value	What it represents	Caution
Municipalities	7	Purposive cases across large/metropolitan and smaller municipalities	Not representative of Italy
Coded lunches	170	Scheduled public-menu lunch days	Provision, not intake
Governance dimensions	12	Standardized public-document review domains	Documentation is not service quality
QA sample	17 rows	Stratified random 10% audit	Roma source version corrected globally
Quantitative outcome cases	3 core cases	Torino, Firenze and Argenta have useful quantitative outcomes	Other cities may collect unpublished data

1. Background and study aim

School meals sit at the intersection of public health nutrition, food procurement, social policy, environmental sustainability and local food systems. Municipal documentation in Italy can contain detailed information on menus, portions, special diets, contract standards, meal commissions, quality controls and tariff systems, but these materials are fragmented across service pages, procurement portals, public-health documents and administrative reports.

The Italian School Meals Observatory – Pilot 2026 was designed to test whether those fragmented public records can be converted into a reproducible comparative monitoring framework. The primary analytical question is not which municipality is “best”. Instead, the study asks what public documents reveal about menu composition, nutritional transparency, governance, monitoring, procurement, equity and measurable outcomes, and where the main evidence gaps remain.

The pilot was intentionally limited to seven municipalities and uses a convenience/purposive design. It should therefore be read as a methodological demonstration and a structured descriptive study, not as an estimate of national school-meal performance.

2. Methods

2.1 Study design

The study is a descriptive cross-sectional public-document assessment. The principal unit of menu analysis is one scheduled school lunch day. Primary-school menus were preferred; when one menu was shared across several school levels, this was explicitly labelled rather than treated as age-specific evidence.

2.2 Sampling of menu cycles

Complete public menu cycles of nine weeks or less were coded in full. Long non-repeating annual calendars were represented by four complete sentinel weeks distributed across the school year. This rule was fixed before final analysis in Codebook v1.0 to limit arbitrary case-by-case sampling decisions. Seasonal menus are treated as season-specific unless multiple seasons are available.

2.3 Coding framework

Binary day-level indicators captured substantial legume dishes, vegetables, fresh fruit, fish, fresh meat, processed meat, eggs, cheese as a main course, explicit wholegrain items, fried/breaded items and plant-based protein main courses. The codebook uses conservative non-inference rules: potatoes do not count as vegetables, fruit juice/purée does not count as fresh fruit, animal ingredients in a first course do not automatically count as a designated meat/fish main, and ambiguous mixed dishes are not decomposed without recipe evidence.

2.4 Missing-data rule

A central methodological rule is that unknown is not zero. Napoli, for example, publicly states fresh seasonal fruit and wholegrain bread provision but does not map those items to specific March 2026 days; those row-level fields remain missing. Argenta does not show fruit/dessert in the public menu table; fruit is therefore also retained as missing rather than coded absent.

2.5 Governance review

Each municipality was reviewed against the same twelve governance/evidence dimensions: public menu availability; recipes/grammages; nutrition/health validation; seasonality; special diets; acceptability monitoring; formal meal-commission/stakeholder mechanisms; waste monitoring/reduction; sustainable procurement; economic access; public outcome data; and control/non-conformity results.

“Documented”, “partial”, “prospective” and “not found in the current review” are evidence-status labels, not quality scores.

2.6 Quality assurance

A fixed-seed stratified random sample of 17 of 170 coded meals was re-checked. The audit deliberately assessed source transcription separately from binary-code consistency. A Roma source-version discrepancy was detected in one sampled row, and all 45 Roma lunches were therefore re-coded from the updated official 2024 winter menu. The codebook itself did not change. After correction, all audited classifications were consistent with Codebook v1.0.

3. Menu composition results

Table 1 summarizes the principal day-level indicators after the Roma QA correction. Frequencies are normalized per five school lunches, except processed meat, which is expressed per 20 lunches. Missing values are left blank where day-level mapping is unavailable.

Municipality	Meals	Vegetables/ wk	Fruit/wk	Legumes/ wk	Fish/wk	Fresh meat/wk	Processed meat/mo	Wholegrain/ wk
Torino	20	4.75	4.00	1.25	1.00	1.00	1.00	0.00
Firenze	20	4.75	4.75	1.50	1.00	1.50	1.00	1.75
Napoli	20	4.50	—	1.00	1.00	1.00	1.00	—
Roma	45	4.33	4.44	1.22	1.00	1.56	0.44	0.56
Guspini	20	5.00	4.00	1.00	1.00	1.25	3.00	0.00
Cittadella	25	4.80	3.00	1.00	0.80	1.80	1.60	1.40
Argenta	20	4.00	—	1.00	1.00	1.75	1.00	0.50

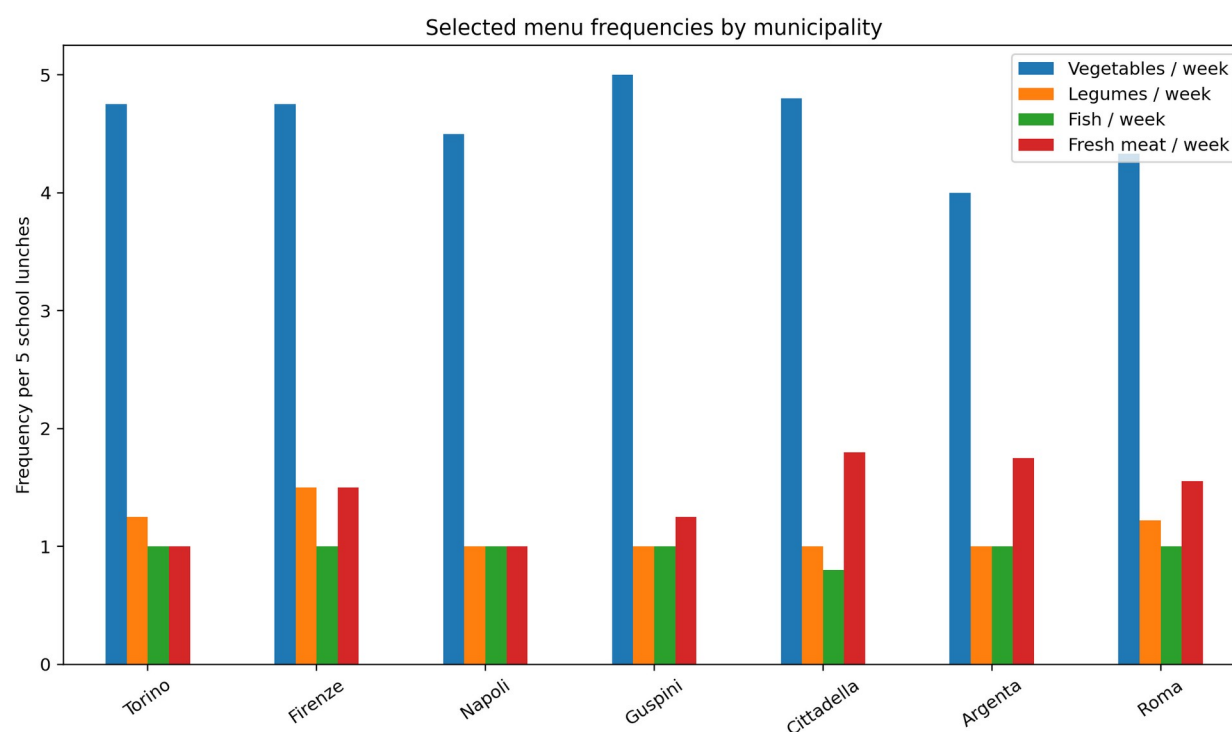


Figure 1. Selected menu frequencies in the seven-municipality pilot. Values are normalized per five coded school lunches.

Fish exposure was notably stable, remaining between 0.8 and 1.0 times per five lunches. Legumes also showed a narrow range of approximately 1.0–1.5 weekly exposures. Vegetable provision was high across all coded menus, from 4.0 to 5.0 days per five lunches. These similarities do not imply equivalent nutritional quality, because dish composition, portions, wholegrain use, preparation method and acceptance differ across systems.

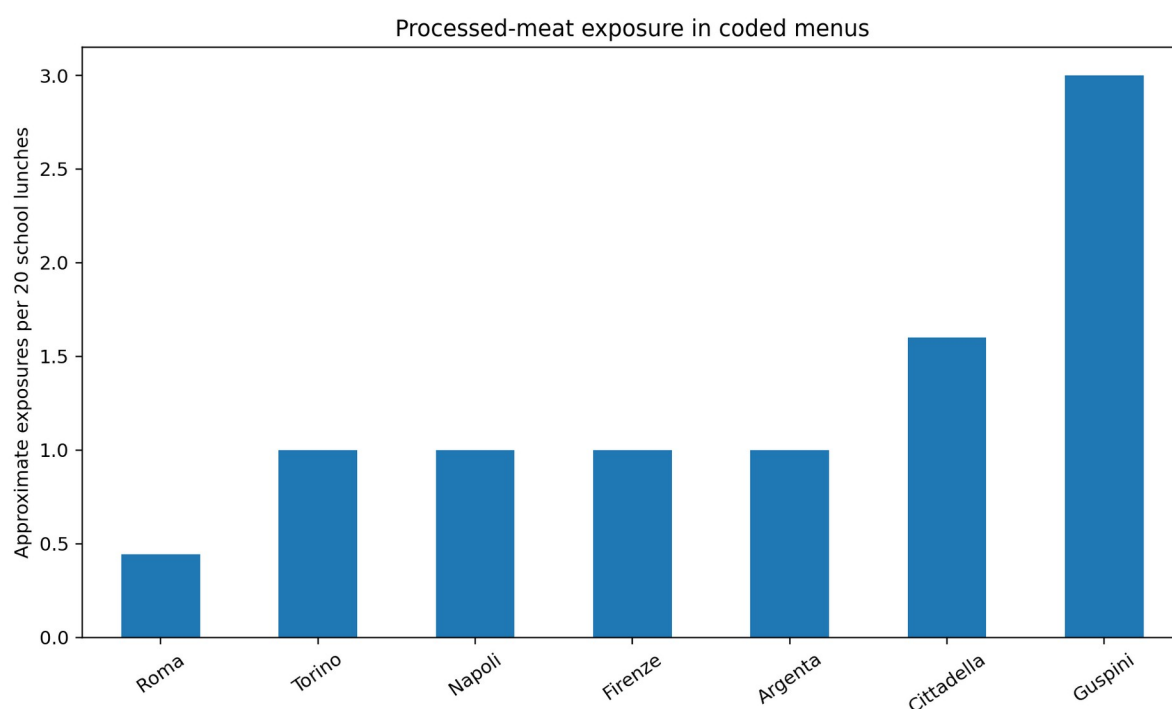


Figure 2. Processed-meat exposure in coded seasonal menus. The measure is an approximate number of exposures per 20 school lunches.

Processed-meat exposure varied more than fish or legumes. Guspini's coded summer cycle had the highest exposure in the pilot, while Roma's corrected winter cycle had the lowest. This is a seasonal menu comparison and should not be interpreted as annual consumption or a municipality quality ranking.

4. Descriptive comparison by municipality size

For exploratory description only, Torino, Firenze, Napoli and Roma were grouped as large/metropolitan cases, while Guspini, Cittadella and Argenta were grouped as smaller municipalities. With only seven purposively selected cases, these averages are not suitable for inferential testing and do not establish a size effect.

Metric	Large / metropolitan	Smaller municipalities
Vegetables / week	4.58	4.60
Fruit / week*	4.40	3.50
Legumes / week	1.24	1.00
Fish / week	1.00	0.93
Fresh meat / week	1.26	1.60
Processed meat / month	0.86	1.87
Wholegrain / week*	0.77	0.63

*Averages ignore missing municipality values. Fruit is unavailable at day level for Napoli and Argenta; wholegrain is unavailable at day level for Napoli.

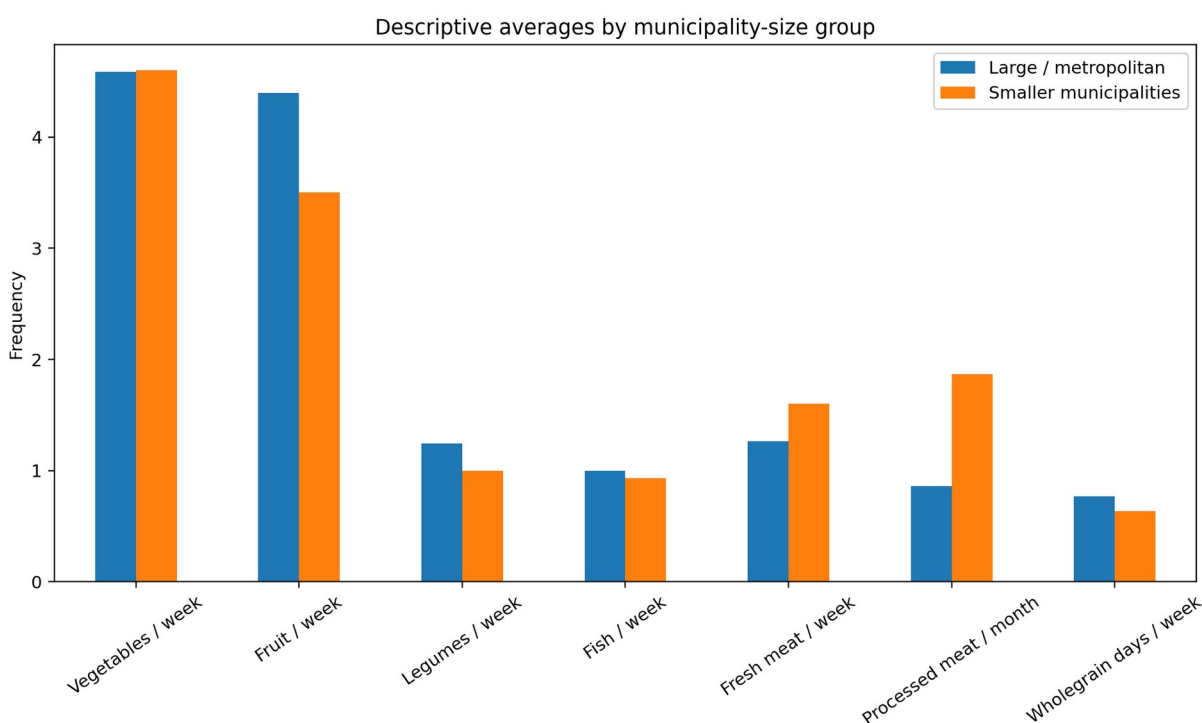


Figure 3. Descriptive group averages. These are exploratory summaries only; the sample is too small and non-random for causal or inferential conclusions.

The most defensible interpretation is that municipality scale does not obviously determine the basic food-group rotation in this small pilot. Vegetables, legumes and fish are similar across the two descriptive groups. Fresh-meat and processed-meat means differ more, but seasonal menu structure and case selection are substantial confounders.

5. Missingness and public-data transparency

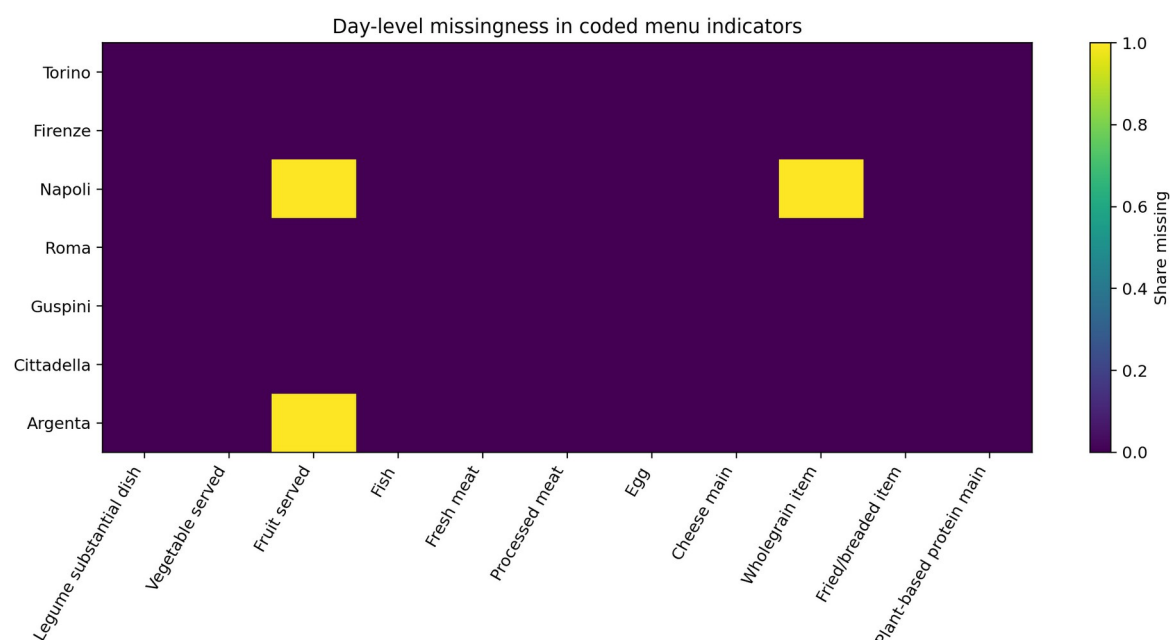


Figure 4. Day-level missingness by municipality and menu indicator. Missingness reflects limitations of public menu documentation, not observed absence of the food item.

Missingness is itself analytically informative. Napoli and Argenta illustrate why a transparency dimension should be kept separate from a nutrition dimension: both services may provide fruit, but the public documents used in the day-level analysis do not allow exact date assignment. Treating these unknowns as zero would mechanically lower their apparent frequency and bias comparisons.

6. Governance, monitoring and procurement

The equal-depth review shows that governance differences are more visible than basic menu differences. All seven municipalities publish menus and all have documented special-diet mechanisms. Formal monitoring and participation systems are also common, but the form and public traceability of those systems differ substantially.

6.1 Torino

Torino combines public menus and grammages with ASL-linked validation, ISO 9001 quality management, formal meal commissions, frequent municipal inspections and a large 2025 customer-satisfaction survey. The principal remaining gaps are quantitative plate-waste results, aggregate non-conformity/sanction data and achieved sustainable-procurement shares.

6.2 Firenze

Firenze is the strongest current multi-domain outcome case in the pilot. It combines public menus and grammages, ASL/USL validation, structured meal commissions, special diets, procurement/CAM documentation and a quantitative 2024 child survey. Public documents also describe waste-monitoring activity, but a citywide quantified plate-waste outcome has not yet been identified.

6.3 Napoli

Napoli's equal-depth review closed an important earlier gap: the city publishes a technical annex with age-specific raw grammages, recipes and detailed merchandise specifications. Governance includes ASL participation, three levels of meal commissions, daily school controls, external monitoring and formal sanctions. Outcome reporting is weaker: satisfaction is described as “sufficient” and external audits as “excellent”, but underlying numeric datasets were not located.

6.4 Roma

Roma is the metropolitan-scale stress test of the pilot, with about 150,000 users per day, more than 30 million annual meals and 730 schools. Daily technical inspections, formal meal commissions, special diets and sustainable sourcing are institutionalized. A citywide 2024 consultation on school meals was conducted, but the detailed result report was not located. The new 48-month procurement remains separate from the observed 2025/26 menu evidence and should only be evaluated as implemented performance after award and service delivery.

6.5 Guspini

Guspini is a useful small-municipality counter-case because it publishes school-level portions and total meal energy and states ASL approval. Current regulation embeds gradimento, special diets, organic/short-chain sourcing and waste reduction. Recent public quantitative satisfaction, waste, control and procurement-performance results remain limited.

6.6 Cittadella

Cittadella is notable for contract design. Its 2025/26–2029/30 specification requires user-satisfaction assessment, measurement of both undistributed food and plate leftovers, semiannual CAM sourcing reports, meal-commission monitoring and a penalty framework. The analytical opportunity is therefore unusually clear: obtain the reports that the contract itself requires and compare intended monitoring with delivered outcomes.

6.7 Argenta

Argenta combines formal parent monitoring with unusually useful administrative outcome data. The 2024 management-control report records 14 schools, 1,046 pupil users and 152,289 meals, while the meal committee can conduct repeated visits and submit structured quality forms. A 2024 school-meal satisfaction survey is explicitly documented, but its detailed result file was not located.

7. Outcome evidence and transparency gap

The most important cross-case limitation is not a lack of policy or monitoring mechanisms; it is a lack of consistently public implementation results. The pilot repeatedly finds that municipalities describe controls, commissions, waste-prevention requirements and sustainable procurement more often than they publish the resulting datasets.

Municipality	Outcome evidence	Current example	Highest-value missing outcome
Torino	Quantitative 2025 parent/user satisfaction	Overall satisfaction 4.84/10; 3,435 valid questionnaires from 15,369 users	Plate waste; sanctions; achieved sourcing
Firenze	Quantitative child survey + service outputs	2024 child survey n=174; additional service/monitoring outputs	Citywide plate waste; achieved sourcing
Napoli	Qualitative public outcomes	Satisfaction reported as sufficient; audits reported as excellent	Underlying numeric satisfaction/audit data; waste; sourcing
Roma	Survey/process evidence	Citywide 2024 consultation documented; detailed results not found	Survey dataset/report; inspection aggregates; waste
Guspini	Process + historical quantitative evidence	Current monitoring process; older inspection records retained only as context	Recent satisfaction, waste, controls, sourcing
Cittadella	Mandatory monitoring outputs	Satisfaction, leftover measurement and CAM reports are contractually required	Actual 2025/26–2026/27 reports
Argenta	Quantitative administrative outcomes + survey existence	2024 service volume/economic outcomes; satisfaction survey documented	Detailed satisfaction scores; current waste/compliance/sourcing

This pattern suggests that a mature version of the Observatory should report two parallel outputs: (1) service characteristics and documented governance, and (2) outcome availability and measured implementation performance. Combining these into one score would obscure important distinctions and reward documentation volume rather than actual performance.

8. Interpretation

Three cross-case messages emerge. First, the seven public menus are less different in core fish, legume and vegetable rotation than a purely anecdotal comparison might suggest. Second, governance systems differ substantially in transparency, monitoring architecture, procurement specificity and stakeholder participation. Third, the main research bottleneck is outcome data: actual waste, dish acceptance, non-conformities and achieved sustainable sourcing are much less consistently published than menu rules and tender requirements.

This matters because a school-food system can look strong on paper yet perform differently in daily implementation. Conversely, a municipality may collect useful outcome data without publishing them in an easily discoverable format. For this reason, the pilot avoids treating documentation depth as a proxy for quality and explicitly separates “required by contract”, “documented process”, “reported outcome” and “quantitative outcome”.

9. Limitations

- Purposive seven-municipality sample; no national representativeness.
- Seasonal non-equivalence: some cases use winter, spring or summer cycles, while Firenze uses four sentinel weeks from a long annual calendar.
- Published menu provision is not equivalent to actual intake, consumption or nutrient adequacy.
- Day-level missingness is driven by public-document structure in some municipalities.
- Procurement requirements do not prove achieved sourcing shares.
- Historical evidence is retained only as historical context and not as current performance.
- Quantitative outcome evidence is uneven, and absence from public search does not prove that data are not collected internally.
- The exploratory alignment percentage remains an internal methodological indicator and should not be used as a public ranking.

10. Next phase

The most valuable next step is targeted retrieval of existing outcome documents rather than immediate expansion to more municipalities. Municipality-specific requests have already been prepared for satisfaction reports, plate-waste data, non-conformity/penalty aggregates, meal-commission records and achieved procurement shares. Requests should explicitly ask for existing records or anonymized datasets and should not require administrations to create new analyses.

While responses are pending, the frozen menu dataset should be analysed reproducibly in R or equivalent code, producing final tables, missingness diagnostics and figures. A second seasonal menu should be added where feasible, particularly for municipalities currently represented by a single seasonal cycle. Expansion to 10–12 municipalities should occur only after the outcome-request and reproducibility stages are stable.

11. Conclusion

The Italian School Meals Observatory – Pilot 2026 demonstrates that publicly available administrative material can be transformed into a traceable cross-municipality school-food monitoring framework. The pilot's main contribution is methodological: it integrates menu composition, transparency, governance, monitoring, procurement and outcome availability without collapsing them into a single quality score.

The current seven-municipality dataset is sufficient to support a credible portfolio-level technical report and to demonstrate applied skills in nutrition, food systems, public procurement analysis, data coding, quality assurance and monitoring/evaluation. The next scientific gain will come from linking those documented systems to real implementation outcomes.

Selected public sources

Torino: Municipal school catering service/menu/quality-control pages and 2025 primary-school customer-satisfaction report. <https://www.comune.torino.it/schede-informative/ristorazione-scolastica-della-citta-torino> ; <https://www.comune.torino.it/media/10527>

Firenze: Municipal school-meal page, ordinary and grammaged menus, meal-commission guidance and 2024 child-satisfaction results. <https://educazione.comune.firenze.it/pagina/mensa>

Napoli: March 2026 menu; technical menu/recipe/merchandise annex; 2025 public-service review; meal-commission regulation.

https://static-www.comune.napoli.it/wp-content/uploads/2025/07/MENU_Ricettario_Tabelle_Merceologiche.pdf

Roma: Municipal school catering service page; updated official winter menu; 2024 consultation notice; procurement portal. <https://www.comune.roma.it/web/it/scheda-servizi.page?contentId=INF802685>

Guspini: Municipal menu, service regulation, monitoring procedures and tariff pages.

<https://www.comune.guspini.su.it/it/page/servizio-mensa-proposta-di-menu>

Cittadella: Municipal school-meal service page and 2025/26–2029/30 tender specifications.

<https://www.comune.cittadella.pd.it/servizio/mensa-scolastica/>

Argenta: Municipal school-meal service, meal committee, 2024 management-control report and 2026 primary menu. <https://www.comune.argenta.fe.it/servizio/iscrizione-servizio-di-refezione>

National reference: Italian Ministry of Health collective-catering guidance; 2020 CAM; FAO School Food Global Hub methodology. <https://www.salute.gov.it/> ; <https://www.gazzettaufficiale.it/eli/id/2020/04/04/20A01905/SG> ; <https://www.fao.org/platforms/school-food/technical-resources/methodology/en>

Appendix A. Reporting rules

- Use “documented”, “reported”, “required by contract”, “prospective” and “not publicly identified in this review” as distinct evidence levels.
- Do not publish an overall municipality ranking.
- Do not convert missing menu fields into zero.
- Do not infer nutrient adequacy from dish names without validated portions/recipes.
- Do not present tender requirements as implemented outcomes.
- For surveys, state respondent numbers and population/denominator when available.
- For historical evidence, label the year explicitly.